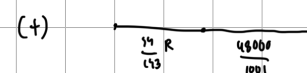
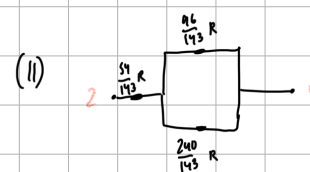
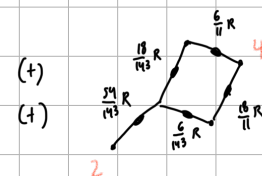
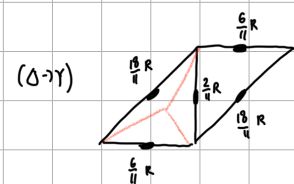
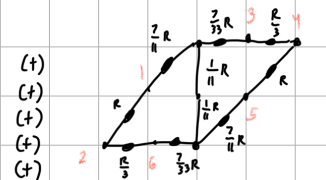
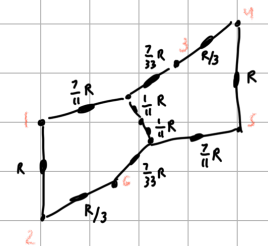
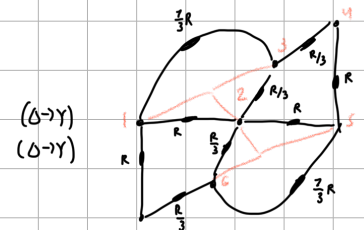
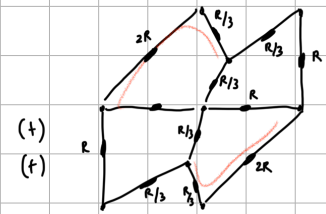
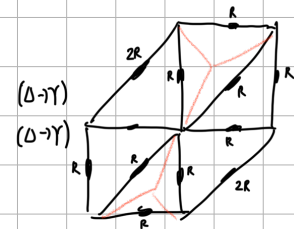
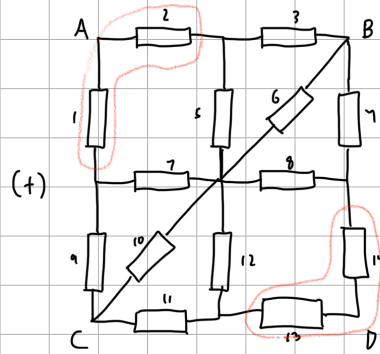


①



$= 85.714 \Omega$

For convenience, we will simplify the circuit using the following "operations":

($\Delta \rightarrow Y$) Delta-Wye

(+) Series

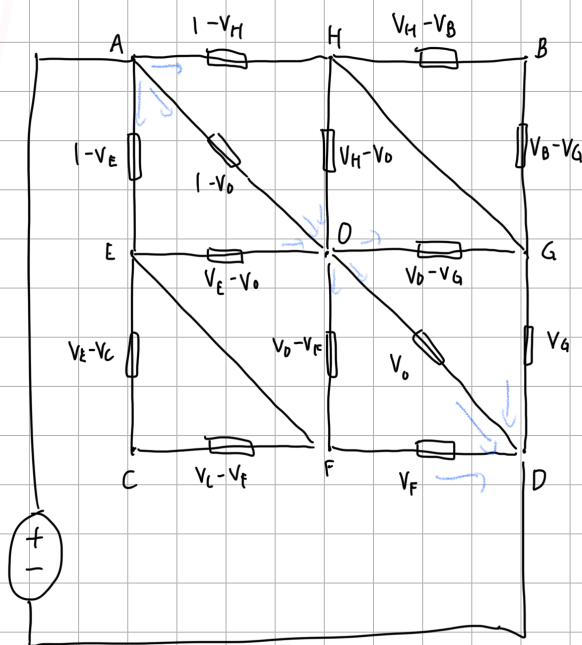
(II) Parallel

(involved resistors are highlighted with red color)

no operation associated implies a redraw for clarity

Multiple operations can happen in one iteration, as long as all values used are from the past iteration and there could be no ambiguity (e.g. the result is the same no matter the order you apply the operations).

2026/09/08



Nodal analysis

$$I_{xy} = \frac{V_x - V_y}{R}$$

Since every current has denominator, let's drop it for the sake of brevity

$$\sum I_{xy} = 0$$

$$\sum \frac{V_x - V_y}{R} = 0$$

$$\sum V_x - V_y = 0$$

Set $V_A := 1 \text{ V}$
 $V_D := 0 \text{ V}$

Note that the circuit is symmetric along $y=x$ and $y=-x$.

That is,

$$V_E = V_H \quad V_C = V_F$$

$$V_G = V_F \quad V_B = V_H$$

$$\Rightarrow V_E = V_G = V_F = V_H \quad V_B = V_C$$

reduce all to V_E

voltage wire

①	$V_F + V_G + V_D$	$= 3 - V_H - V_E - V_D$
D	② $(1 - V_D) + (V_H - V_D) + (V_E - V_D)$	$= (V_D - V_G) + (V_D) + (V_D - V_F)$
G	③ $(V_B - V_G) + (V_D - V_G)$	$= V_G$
F	④ $(V_C - V_F) + (V_D - V_F)$	$= V_F$

reduce all to V_E

①	$2V_E + V_D$	$= 3 - 2V_E - V_D$
②	$1 + 2V_E - 3V_D$	$= 3V_D - 2V_E$
③	$V_B + V_D - 2V_E$	$= V_E$
④	$V_C + V_D - 2V_F$	$= V_F$

Hence, current is

$$I_A = \frac{(V_A - V_H) + (V_H - V_E) + (V_H - V_D)}{R}$$

$$= \frac{\frac{1}{2} + \frac{1}{2} + \frac{1}{2}}{R}$$

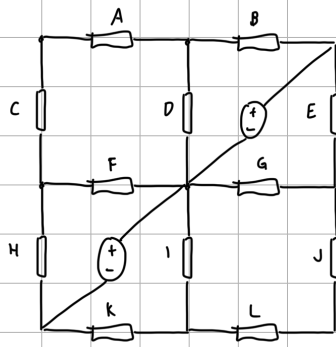
$$I_A = \frac{3}{2R}$$

$$R_{AD} = \frac{1-0}{I_A} = \frac{1}{1.5/R} = \frac{2}{3}R = \boxed{133.333 \Omega}$$

①	$4V_E + 2V_D$	$= 3$
②	$-4V_E + 6V_D$	$= 1$

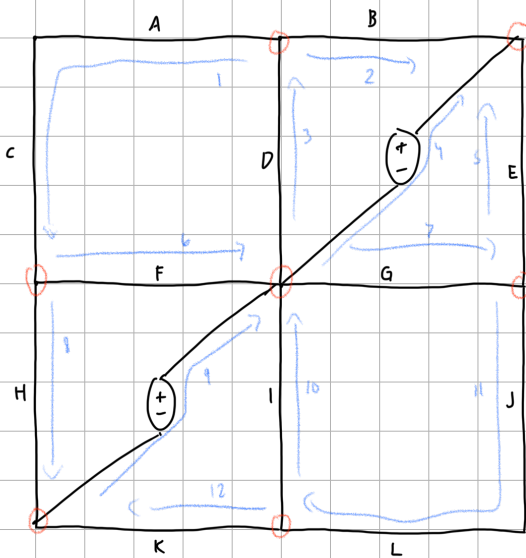
$$\Rightarrow \begin{matrix} 4V_D \\ V_D \end{matrix} = \begin{matrix} 4 \\ 1/2 \end{matrix}$$

so $\begin{matrix} 4V_E + 1 \\ 4V_E \\ V_E \end{matrix} = \begin{matrix} 3 \\ 2 \\ 1/2 \end{matrix} = V_G = V_F = V_H$



1 2 3 4 5 6 7 8 9 10 11 12 V

KJR



$$I_1 = I_6 + I_8$$

$$I_4 + I_2 + I_3 = 0$$

$$I_8 + I_{12} = I_9$$

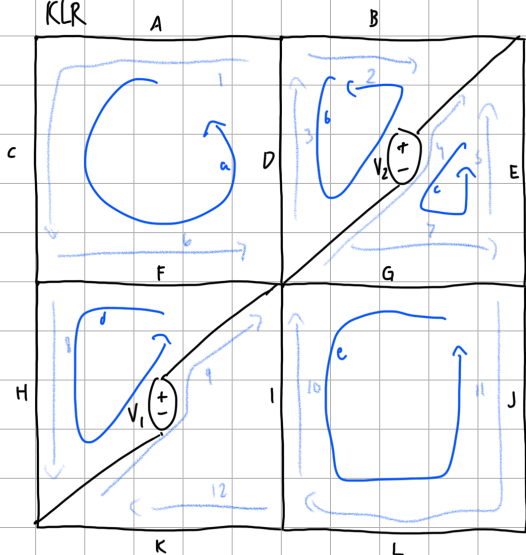
$$I_{11} = I_{10} + I_{12}$$

$$I_7 = I_5 + I_{11}$$

$$I_3 = I_1 + I_2$$

$$I_9 + I_6 + I_{10} = I_3 + I_4 + I_7$$

KLR



$$(a) -I_2 R_A - I_6 R_F - I_3 R_D = 0$$

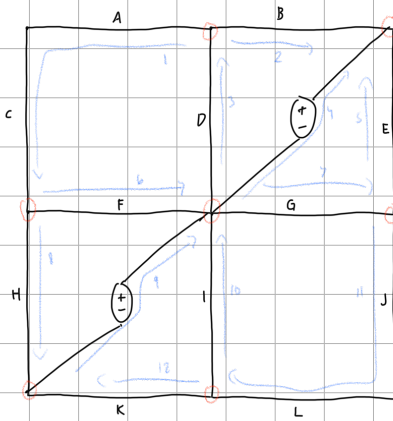
$$(b) +V_2 + I_2 R_B + I_3 R_D = 0$$

$$(c) -I_7 R_G - I_5 R_E - V_2 = 0$$

$$(d) -I_8 R_H + V_1 + I_6 R_F = 0$$

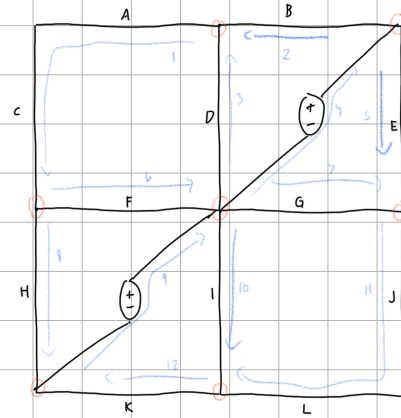
$$(e) +I_{10} R_I + I_{11} R_L + I_{11} R_J + I_7 R_G = 0$$

KJR

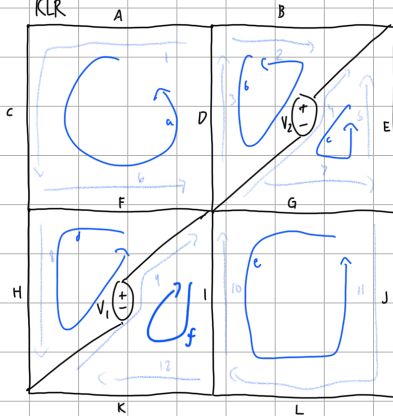


$$\begin{aligned} I_1 &= I_6 + I_8 \\ I_4 + I_2 + I_5 &= 0 \\ I_8 + I_{12} &= I_9 \\ I_{11} &= I_{10} + I_{12} \\ I_7 &= I_5 + I_{11} \\ I_3 &= I_1 + I_2 \\ I_9 + I_6 + I_{10} &= I_3 + I_4 + I_7 \end{aligned}$$

Corrected Directions:



KLR



$$\begin{aligned} (a) -I_1(R_A + R_C) - I_6 R_F - I_3 R_D &= 0 \\ (b) +V_2 + I_2 R_B + I_3 R_D &= 0 \\ (c) -I_7 R_G - I_5 R_E - V_2 &= 0 \\ (d) -I_8 R_H + V_1 + I_6 R_F &= 0 \\ (e) +I_{10} R_J + I_{11} R_L + I_{12} R_K + I_7 R_G &= 0 \\ (f) -I_{12} R_K + V_1 + I_{10} R_J &= 0 \end{aligned}$$

$$\begin{matrix} R & A, C & B & D & V & E & F & G & H & V & I & J, L & K \\ I & 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 & 9 & 10 & 11 & 12 & V \end{matrix}$$

$$\begin{matrix} 1 & \left[\begin{array}{cccccccccccc} & & & & & & -1 & & -1 & & & & \\ & & & & & & & & & & & & \\ 2 & & & & & & & & & & & & \\ 3 & & & & & & & & & & & & \\ 4 & & & & & & & & & & & & \\ 5 & & & & & & & & & & & & \\ 6 & & & & & & & & & & & & \\ 7 & & & & & & & & & & & & \\ 8 & & & & & & & & & & & & \\ 9 & & & & & & & & & & & & \\ 10 & & & & & & & & & & & & \\ 11 & & & & & & & & & & & & \\ 12 & & & & & & & & & & & & \\ 13 & & & & & & & & & & & & \end{array} \right] \end{matrix}$$

$$\begin{aligned} I_1 &= I_6 + I_8 \\ I_4 + I_2 + I_5 &= 0 \\ I_8 + I_{12} &= I_9 \\ I_{11} &= I_{10} + I_{12} \\ I_7 &= I_5 + I_{11} \\ I_3 &= I_1 + I_2 \\ I_9 + I_6 + I_{10} &= I_3 + I_4 + I_7 \end{aligned}$$

3 - R with highest current

4 - I

5 - R with lowest current

6 - I

7 - R with highest voltage drop

8 - V

9 - R with lowest voltage drop

10 - V

Plugging to an RREF solver (code attached):

Using Ohm's law

$$V_x = I_x R_x$$

	I	(mA)
Arc	1	136.363
B	2	-130.303
D	3	6.060
	4	251.732
E	5	-121.428
F	6	-60.606
G	7	7.142
H	8	196.969
	9	419.398
I	10	-92.857
J, L	11	128.571
K	12	221.428

Just means direction
assigned was wrong.
Take absolute value for
current answer.

V	(V)
A	1.363
B	1.303
C	1.363
D	0.303
E	1.214
F	3.030
G	0.214
H	1.969
I	2.785
J	1.285
K	2.214
L	1.285

- 3 - R with highest current
- 4 - I
- 5 - R with lowest current
- 6 - I
- 7 - R with highest voltage drop
- 8 - V
- 9 - R with lowest voltage drop
- 10 - V

K	221.428 mA
D	6.060 mA
F	3.030 V
G	0.214 V

- ① By KVL, the total voltage across series is sum of voltage drops, while current is constant.

$$V_{\text{tot}} = \sum V_i$$

$$I = I_1 = I_2 = \dots = I_n$$

Note that $V = IR$

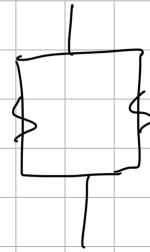
$$\text{so } I_{\text{tot}} R_{\text{tot}} = \sum I_i R_i$$

$$I R_{\text{tot}} = \sum I R_i = I \sum R_i$$

$$R_{\text{tot}} = \sum R_i$$

This generalizes to n resistors using induction.

- ② Note that parallel components have the same voltage



$$V = V_1 = V_2$$

$$I R_1 = I R_2$$

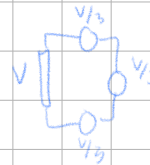
$$\text{so } I_1 = \frac{V}{R_1} \quad I_2 = \frac{V}{R_2}$$

while current splits

$$\text{By KCL: } I = I_1 + I_2$$

$$\frac{V}{R_{\text{eq}}} = \frac{V}{R_1} + \frac{V}{R_2}$$

$$\frac{1}{R_{\text{eq}}} = \frac{1}{R_1} + \frac{1}{R_2} \Rightarrow R_{\text{eq}} = \left(\frac{1}{R_1} + \frac{1}{R_2} \right)^{-1}$$



Solution:

Start from $n/3$:



split this R into



split this $2R$ into



Our smallest resistor is $\frac{4R}{4}$.

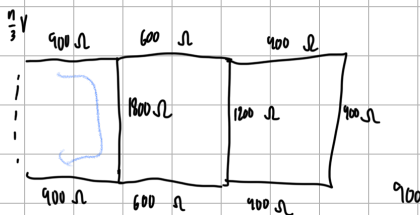
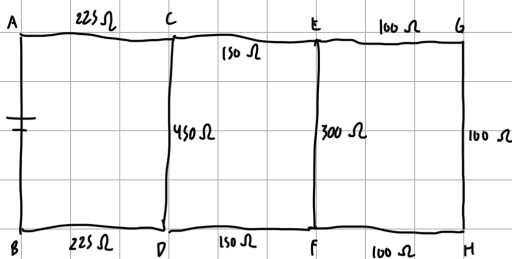
$$\text{so } \frac{4R}{4} = 100 \Rightarrow R = \frac{900}{4} = 225$$

$$\frac{2R}{3} = \frac{225(2)}{3} = 150$$

Hence:

$\frac{n}{3}$	AC, BD
$\frac{n}{4}$	CE, DF
$\frac{n}{27}$	EG, FH, FI

③



900Ω

$$6 \rightarrow 2 \rightarrow$$

$$2.25$$

$$1.5$$

$$1$$

$$9 \rightarrow 4.5 \rightarrow 2.25$$

$$4.5$$

$$3$$

$$1$$

$$7.5 \rightarrow 3.7$$

$$2.25$$

$$1.5$$

$$1$$

$$6$$